

Name	Definition	Type
Amazon Athena	Amazon Athena is a fully managed, serverless query engine that enables SQL-based analysis of structured, semi-structured, and unstructured data stored in Amazon S3. It leverages Presto, an open-source distributed SQL query engine, to perform fast, parallelized queries across large datasets without the need to provision or manage servers. Athena supports a variety of data formats including CSV, JSON, Parquet, ORC, and Avro, and integrates with AWS Glue Data Catalog for schema discovery, metadata management, and partitioning. Athena supports ANSI SQL queries, including joins, complex aggregations, and window functions, and can efficiently query large datasets by leveraging partitioning and columnar formats (like Parquet or ORC) to minimize data scans and reduce query costs. It automatically scales based on query complexity and workload, dynamically optimizing query execution plans. Athena charges based on the amount of data scanned per query, encouraging best practices such as data partitioning and use of columnar formats for cost efficiency. It can also be integrated with AWS services like Lambda, QuickSight, and CloudWatch for extended analytics and automation, making it suitable for real-time log analysis, ad-hoc reporting, and big data workflows.	Analytics-Technology
Amazon Data Firehose	Amazon Kinesis Data Firehose is a fully managed, serverless service for streaming data ingestion and delivery. It buffers, compresses (GZIP, Snappy), and encrypts data before delivering it to destinations like Amazon S3, Redshift, Elasticsearch, or custom HTTP endpoints. Firehose supports real-time data transformation using AWS Lambda functions, allowing for on-the-fly processing. It automatically scales to handle variable throughput without user intervention, with fine-grained control over buffering and delivery intervals. Firehose is optimized for high-throughput, low-latency streaming use cases and charges based on data volume ingested, processed, and delivered.	Analytics-Technology
Amazon EMR	Amazon EMR (Elastic MapReduce) is a cloud-native big data platform that simplifies processing vast amounts of data using open-source tools like Apache Hadoop, Spark, HBase, and Hive. It allows users to quickly deploy, manage, and scale clusters for data analytics, machine learning, and other big data workloads. EMR integrates with other AWS services (e.g., S3, DynamoDB, Redshift) and provides flexibility in cluster configuration and scaling, making it suitable for both batch and real-time processing. It is fully managed, with automated provisioning, scaling, and maintenance, reducing operational complexity and cost.	Analytics-Technology
Amazon Glue	AWS Glue is a fully managed extract, transform, and load (ETL) service designed for preparing and transforming data for analytics. It automates data discovery, schema recognition, and job scheduling, simplifying data integration across a variety of sources (e.g., Amazon S3, RDS, Redshift). AWS Glue provides tools like Glue Data Catalog for metadata storage, Glue Crawlers for data discovery, and Glue Studio for visual ETL job creation. It supports both serverless and provisioned environments, enabling scalable, cost-efficient data workflows. Glue can be integrated with other AWS services and is optimized for big data processing with Apache Spark.	Analytics-Technology
Amazon Kinesis	Amazon Kinesis is a suite of services designed to handle real-time data streaming and processing at scale. It allows you to collect, process, and analyze large amounts of data from various sources like IoT devices, logs, and social media in real time. Kinesis consists of several components: Kinesis Data Streams: Captures and stores large streams of data, enabling real-time processing and analytics. Data is ingested in shards, which can be scaled based on throughput needs. Kinesis Data Firehose: Automatically loads streaming data into other AWS services like Amazon S3, Redshift, or Elasticsearch for further analysis, without needing custom code. Kinesis Data Analytics: Provides real-time SQL querying and analytics on streaming data, enabling users to process data as it arrives and generate actionable insights. Kinesis Video Streams: Ingests, stores, and processes video streams for use cases like surveillance, media, and machine learning applications. Kinesis offers scalability, low-latency processing, and integration with other AWS services, making it a powerful tool for managing real-time data pipelines.	Analytics-Technology

Amazon Kinesis Data Streams	Amazon Kinesis Data Streams is a scalable and durable service for real-time streaming data collection and processing. It allows you to ingest large volumes of data, such as logs, social media feeds, or telemetry data, in real time. The service breaks the data into shards, which are the basic units of capacity and parallelism. You can consume the data using Kinesis Data Streams APIs or integrate with other AWS services like Kinesis Data Analytics and Kinesis Data Firehose for further processing and storage. Kinesis Data Streams offers features like data retention, automatic scaling, and fault tolerance, making it suitable for high-throughput, low-latency streaming applications.	Analytics-Technology
AWS Lake Formation	AWS Lake Formation is a managed service that simplifies building and securing a data lake on Amazon S3. It leverages AWS Glue for ETL, metadata cataloging, and schema management. Lake Formation provides fine-grained access control, enabling centralized permissions at the table or column level using IAM and KMS integration. It uses the AWS Glue Data Catalog as the central metadata store. Security features include encryption (at rest and in transit), audit logging with CloudTrail, and cross-account data sharing. It integrates natively with analytics services like Athena, Redshift Spectrum, and SageMaker for scalable, secure data processing.	Analytics-Technology
Amazon Managed Service for Apache Flink	Amazon Managed Service for Apache Flink is a fully managed platform for real-time stream and batch processing with Apache Flink. It abstracts cluster management, auto-scaling, and fault tolerance, while supporting stateful processing with checkpointing. The service integrates with data sources (e.g., Kinesis, Kafka, S3) and sinks (e.g., Redshift, Elasticsearch). It offers seamless scaling, high availability, and secure execution within a VPC with IAM and encryption support. This service streamlines Flink job deployment and operation, providing simplified management for complex data processing workflows.	Analytics-Technology
Amazon OpenSearch Service	Amazon OpenSearch Service is a fully managed service for running OpenSearch clusters, providing real-time search and analytics on large datasets. It supports full-text search, log analytics, and metrics visualization. Key features include auto-scaling, automated backups, fine-grained access control, and encryption (at rest and in transit). It integrates with AWS data sources (e.g., Kinesis, S3) and visualization tools like Kibana, enabling scalable, low-latency search and analytics workloads.	Analytics-Technology
Amazon QuickSight	Amazon QuickSight is a managed BI service for interactive dashboards and data visualizations. It integrates with AWS data sources (e.g., S3, RDS, Redshift) and uses the SPICE engine for fast in-memory processing. Features include embedded ML insights (anomaly detection, forecasting), auto-scaling, fine-grained access control, and encryption. QuickSight supports embedding and sharing dashboards, providing scalable and secure BI with minimal infrastructure.	Analytics-Technology
AWS Batch	AWS Batch is a fully managed service for running batch jobs at scale. It dynamically provisions compute resources (EC2, Spot, Fargate) based on job requirements. Key features include job scheduling, prioritized queues, and integration with AWS services (e.g., S3, Lambda, CloudWatch). It optimizes cost and performance, supporting EC2 Spot for cost-effective execution.	Compute-Technology
Amazon EC2	Amazon EC2 (Elastic Compute Cloud) is a scalable compute service that provides resizable virtual machines (instances) on-demand. It supports multiple instance types for various workloads, with customizable CPU, memory, storage, and networking. EC2 integrates with other AWS services (e.g., S3, VPC, EBS) and supports auto-scaling, load balancing, and security configurations via IAM. It offers flexible pricing models, including On-Demand, Reserved, and Spot Instances, enabling cost optimization based on usage patterns.	Compute-Technology
AWS Lambda	AWS Lambda is a serverless compute service that automatically runs code in response to events without provisioning or managing servers. It supports multiple languages (e.g., Python, Node.js, Java) and integrates with AWS services like S3, DynamoDB, and API Gateway. Lambda automatically scales based on incoming requests and charges only for the compute time used. It enables event-driven architectures and offers fine-grained permissions through IAM for security.	Compute-Technology

Amazon Elastic Container Registry (ECR)	Amazon Elastic Container Registry (ECR) is a fully managed Docker container registry that stores, manages, and deploys container images. It integrates with Amazon ECS, EKS, and AWS Lambda, providing secure image storage with encryption at rest (via KMS) and in transit. ECR supports image versioning, lifecycle policies, and fine-grained access control through IAM. It also includes automated image scanning for vulnerabilities and is optimized for high availability and scalability.	Containers-Technology
Amazon Elastic Container Service (ECS)	Amazon Elastic Container Service (ECS) is a fully managed container orchestration service for running Docker containers at scale. It supports both EC2 and Fargate launch types for compute, allowing users to choose between managing their own instances or using serverless compute. ECS integrates with other AWS services like IAM, VPC, and CloudWatch for security, networking, and monitoring. It offers features like task definitions, service discovery, and auto-scaling for efficient containerized application management.	Containers-Technology
Amazon Elastic Kubernetes Service (EKS)	Amazon Elastic Kubernetes Service (EKS) is a fully managed service for running Kubernetes clusters on AWS. It automates cluster management tasks such as patching, scaling, and securing the Kubernetes control plane. EKS integrates with AWS services like IAM, VPC, and CloudWatch for security, networking, and monitoring. It supports both EC2 and Fargate compute options for running containerized workloads and offers features like automatic scaling, service discovery, and high availability.	Containers-Technology
AWS Fargate	AWS Fargate is a serverless compute engine for running containers without managing servers or clusters. It integrates with Amazon ECS and EKS, allowing users to run Docker containers with just the application code and resource specifications. Fargate automatically provisions and scales compute resources based on workload requirements. It supports fine-grained IAM permissions, integrates with VPC for networking, and provides built-in security features like encryption at rest and in transit.	Containers-Technology
Amazon Redshift	Amazon Redshift is a fully managed, petabyte-scale data warehouse service optimized for OLAP workloads. It uses columnar storage, data compression, and parallel query execution to enable high-performance analytics. Redshift employs massively parallel processing (MPP) to distribute queries across multiple nodes for fast query execution. It supports SQL-based querying with compatibility to PostgreSQL, and integrates with data lakes, AWS Glue, and Amazon S3. Redshift Spectrum allows direct querying of data in S3 without loading it into the cluster. The service offers automatic scaling, data encryption (at rest and in transit), and fine-grained access control via IAM. Redshift's workload management (WLM) helps optimize query performance, and the use of machine learning models enables intelligent query optimization.	Database-Technology
AWS IoT Greengrass	AWS IoT Greengrass is a service that extends AWS IoT capabilities to local devices, enabling edge computing. It allows devices to run AWS Lambda functions, manage data, and interact with local resources (e.g., sensors, actuators) even without a constant internet connection. Greengrass supports secure device communication with AWS, data stream processing, and local data storage. It provides features like over-the-air (OTA) updates, device shadows, and encryption for data at rest and in transit, enabling scalable and reliable IoT applications at the edge.	InternetOfThings-Technology
Amazon Bedrock	Amazon Bedrock is a fully managed service for building and scaling generative AI applications using pre-trained foundation models (FMs) from leading AI providers like Anthropic, Stability AI, and OpenAI. It simplifies integration with FMs for tasks such as natural language processing, image generation, and code completion, offering low-latency inference and customizability through fine-tuning. Bedrock provides an easy-to-use API, secure access with IAM, and scalability, enabling organizations to deploy AI-driven solutions without managing the underlying infrastructure.	MachineLearning-Technology
Amazon Comprehend	Amazon Comprehend is a fully managed natural language processing (NLP) service that uses machine learning to extract insights from text. It supports tasks like sentiment analysis, entity recognition, language detection, and topic modeling. Comprehend offers custom entity recognition and classification models, and integrates with AWS services like S3 and Lambda. It uses advanced deep learning algorithms for real-time text analysis, and provides automatic scaling, with data security through encryption at rest and in transit.	MachineLearning-Technology

AWS Deep Learning AMIs (DLAMI)	AWS Deep Learning AMIs (DLAMI) are pre-configured Amazon Machine Images designed to accelerate deep learning model development. They come with popular frameworks like TensorFlow, PyTorch, and MXNet, as well as NVIDIA CUDA, cuDNN, and other libraries optimized for GPU instances. DLAMIs support both CPU and GPU-based EC2 instances, providing flexibility for training and inference. They also integrate with AWS services like S3, EFS, and SageMaker for data storage and model management. DLAMIs enable rapid deployment of custom deep learning workflows while offering security and scalability.	MachineLearning-Technology
Amazon Forecast	Amazon Forecast is a fully managed service that uses machine learning to generate accurate time-series forecasts. It automatically builds, trains, and deploys models by leveraging historical data, providing capabilities like demand forecasting, resource planning, and financial forecasting. Forecast integrates with AWS data sources like S3, RDS, and Redshift and supports custom datasets for enhanced accuracy. It offers built-in anomaly detection, seasonal adjustments, and integrates with other services like SageMaker for advanced model customization. The service automatically scales and handles model optimization, with data security via encryption at rest and in transit.	MachineLearning-Technology
Amazon Fraud Detector	Amazon Fraud Detector is a fully managed service that uses machine learning to detect and prevent fraud in real-time transactions. It automates model creation and training using historical data to identify patterns and anomalies associated with fraudulent activities. The service supports custom fraud detection use cases, such as payment fraud, account takeovers, and loyalty fraud, with features like feature engineering, model versioning, and real-time scoring. Fraud Detector integrates with other AWS services like Lambda and S3 for data processing and automated workflows, and ensures security through encryption and access control via IAM.	MachineLearning-Technology
Amazon Lex	Amazon Lex is a fully managed service for building conversational interfaces using voice and text. It enables the creation of chatbots and virtual assistants by leveraging automatic speech recognition (ASR) and natural language understanding (NLU) to interpret user input. Lex supports integration with AWS Lambda for backend logic and other AWS services like S3, DynamoDB, and CloudWatch. It provides intent recognition, slot filling, and dialog management, and scales automatically based on usage. Lex also supports multi-turn conversations, security through IAM, and encryption at rest and in transit.	MachineLearning-Technology
Amazon Kendra	Amazon Kendra is a fully managed, intelligent search service powered by machine learning, designed to deliver highly accurate, context-aware search results. It uses natural language processing (NLP) to understand user queries and retrieve relevant information from a variety of data sources, including S3, SharePoint, Salesforce, and databases. Kendra provides built-in connectors, document ranking, and supports custom relevance tuning for improved search accuracy. It integrates with AWS services like Lambda for custom workflows, IAM for access control, and offers security with encryption at rest and in transit.	MachineLearning-Technology
Amazon Mechanical Turk	Amazon Mechanical Turk (MTurk) is a fully managed crowdsourcing service that enables businesses to offload tasks requiring human intelligence, such as data labeling, content moderation, and surveys. It connects businesses with a global workforce to complete micro-tasks through a web-based interface. MTurk supports task management, quality control through HITs (Human Intelligence Tasks), and worker performance tracking. It integrates with AWS services like S3, Lambda, and Step Functions for automation and data processing, offering flexible pricing and scalability. MTurk ensures security and compliance through IAM-based access control and encryption.	MachineLearning-Technology
Amazon Polly	Amazon Polly is a fully managed text-to-speech (TTS) service that converts text into lifelike speech using deep learning models. It supports multiple languages and voices, including neural network-based voices for improved naturalness and expressiveness. Polly integrates with AWS services like Lambda and S3 for real-time or batch processing of text, and allows customization of speech output with SSML (Speech Synthesis Markup Language). It provides features like speech marks, lexicons, and streaming audio, and ensures security through IAM access control and encryption at rest and in transit.	MachineLearning-Technology
Amazon Q		